

Standardization and ITU

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Fireworks meeting
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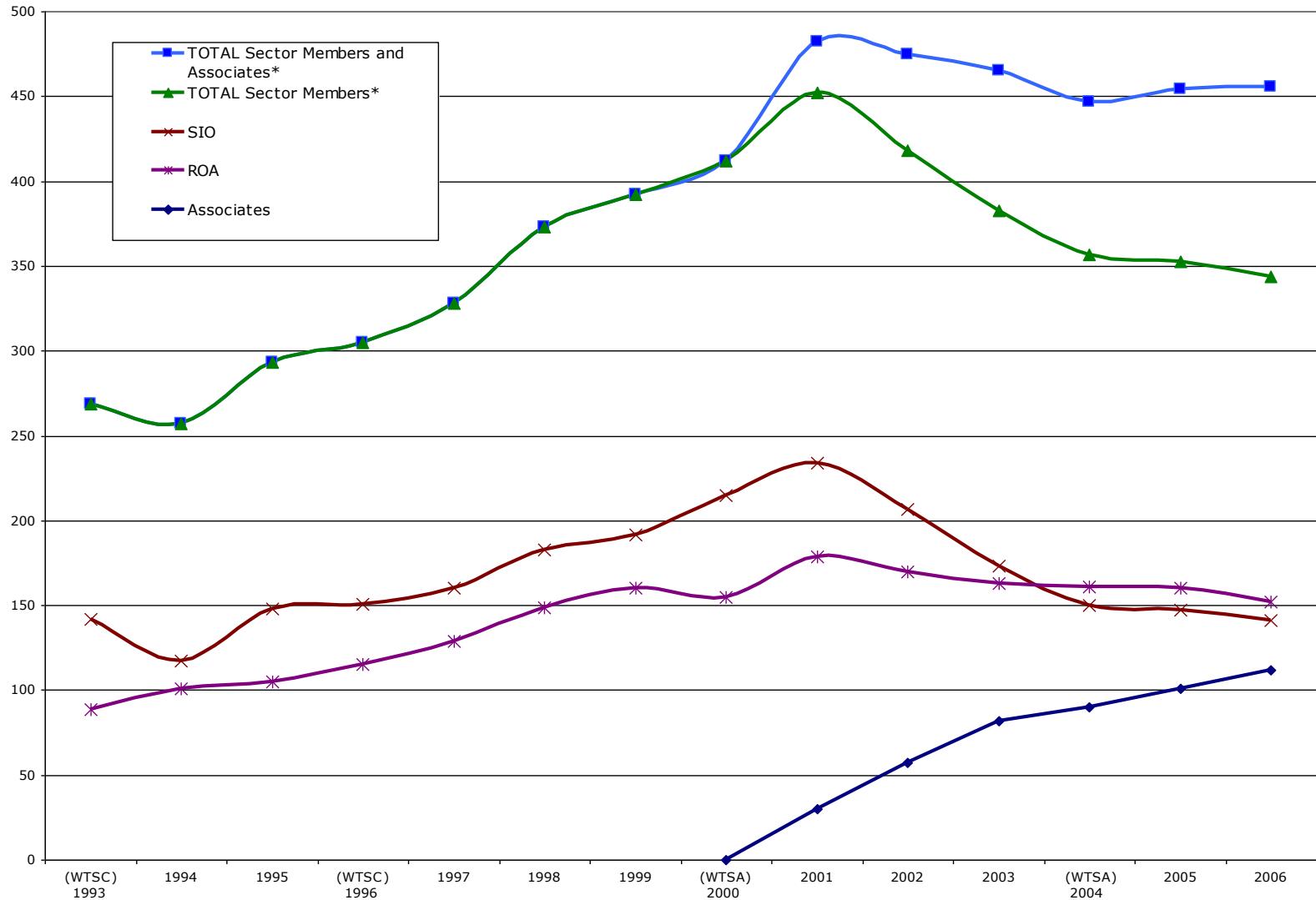
Outline

- ITU-T Characteristics
- Work highlights
- Focus Groups
- Cooperation

ITU-T Characteristics

- Market pressure forced, and continues to force ITU-T to adapt
- Unique partnership of private sector and government
- Work (mostly) done in Study Groups (13 of them)
- Today, 95% of work is done by private sector
 - Strong participation from Asia, esp. China
- Truly global
- Consensus decisions
- Very flexible
- Fast procedures, transparent procedures
 - start work: 1 day / few weeks
 - develop work: weeks to 2-3 yrs
 - approve work: average 2 months
 - ▶ AAP for technical standards (= 95% of work)
 - publish work: couple of days after approval
- IPR Policy

Sector Members and Associates



* including regional and other international organizations, regional telecommunications organizations, intergovernmental organizations and other entities dealing with telecommunications

Top-13 SG and WP Participation (2006)

= 79% of total
admin attendance

76%

69%

49%

Administrations (841/1067) Members: 191		ROA (328/432) Members: 154		SIOs (700/1016) Members: 141		Associates (42/86) Members: 112	
China	153	France Telecom	60	NTT *	129	HHI	6
Korea	153	BT	47	Huawei	123	Dilithium	5
United States	132	KDDI	36	ETRI	90	OFS Fitel	4
United Kingdom	68	China Telecoms.	35	Nortel Networks *	53	Semtech	4
Syria	62	Telekom.Polska	26	Siemens *	48	UpZide	4
Japan	49	Deutsche Telekom	20	Alcatel *	47	AULM	3
France	44	Telecom Italia	18	Lucent *	47	RAD Data	3
Russia	44	AT&T	17	ZTE	45	Yokogawa	3
Canada	40	China Mobile Com.	16	Cisco	33	Anagran	2
Germany	35	Swisscom	16	L.M. Ericsson	23	ASSIA	2
Australia	25	NHK	13	NEC	21	Atos Origin	2
Switzerland	20	Telenor ASA	13	Telcordia	21	Essex	2
Kenya	16	KT Corp.	11	Hitachi	20	Finisar	2

Number of
participations

Sector Members

Associates



* Consolidated for various memberships

Work highlights

ITU-T hot topics

- Next generation network (NGN)
- IPTV
- Cybersecurity including identity management (IdM)
- Ubiquitous networks
- Next generation multimedia conferencing
- Videocoding
- Broadband access
- Packet based transport
- Fibre optics
- Home networking
- ICTs and climate change

Work highlights 2006

- 231 Recs/supplements published
- Increase of 30% over 2005 in terms of meeting documents processed
- 61 meetings (44 in Geneva, 17 outside)
 - Increase of 13% over 2005
- Strong NGN Focus
 - Global NGN standards progressing rapidly within ITU-T
 - Providing global leadership through overall framework and structure
 - Building on the work of other bodies (not reinventing ...)
 - 22 Recommendations on NGN
 - Transition from legacy, requirements, architecture, security, QoS, OAM (Ethernet and MPLS)

Upcoming workshops

- **Making accessibility a reality in emerging technologies:** 13 November, Rio de Janeiro, Brazil
- **Human exposure to electromagnetic fields (EMFs)**
20 November, Geneva, Switzerland
- **Broadband Wireless Access Seminar**
26 November, Moscow, Russia
- **Standardization of mechanisms for ensuring ICT security and development of ICT services**
6 December, Belarus
- **The Fully Networked Car**
7 March 2008 - PALEXPO, Geneva, Switzerland

Innovations in NGN

12-13 May 2008

- Part of initiative to increase dialogue between academia/ITU
- Follows January 2007 consultation meeting
- First in a series of '**Kaleidoscope**' events
- Will identify new topics for standardization
- Call for papers issued
 - Deadline 15 October
- itu.int/ITU-T/uni/kaleidoscope/

Bridging the standardization gap: Forums 2008

- Addressing manufacturers, operators, service providers, regulators and administrations.
- Providing an overview of technologies that have created major standardization challenges such as NGN, multimedia, VoIP, IPTV, security and regulation.
- Highlighting ways and means to enhance cooperation and participation in ITU's standardization work and standards implementation in developing countries.
- One Forum in each region – Asia Pacific, Americas, Arab, Africa – before October 2008

What is the standardization gap?

- The standardization gap might be defined as disparities in the ability of representatives of developing countries, relative to developing ones, to **access, implement, contribute to and influence international ICT standards**, specifically ITU Recommendations.
- The standardization gap is itself both **a cause and a manifestation of the wider digital divide**
- It contributes to the ***persistence*** of the wider digital divide

What has been done recently to reduce the standardization gap?

- First in series of annual Forums in each region organised by TSB, BDT, Regional Office and Regional Organisation
- Placing greater emphasis on implementation guidelines – exploring possible translation in regions?
- Starting a series of Tutorial Groups on implementation
- Directors TSB and BDT written to all Member States and Sector Members inviting contributions for fund on bridging standardization gap
- Establishing Regional Groups (e.g. SG2 for Arab Region)
- Trialling collaborative tool for remote participation in meetings

Focus Groups

ITU-T Focus Groups

- Augment Study Group work providing quick reaction to standardization needs and giving participation and working method flexibility
 - Forum-like entities with “arms-length” organization under ITU-T Study Group
 - A high degree of independence
 - Outputs in well-defined areas:
 - Stand-alone Focus Group deliverables or
 - Provided as direct input into Study Groups for progression to ITU-T Recommendation
 - Non-ITU members can participate and can benefit from:
 - Worldwide visibility
 - Networking opportunities
 - Exposure to a large pool of expertise drawn from related work under progress in ITU-T Study Groups and other ITU-T Focus Groups.

ITU-T Focus Groups (1)

■ **Identity Management**

Meeting 1: Geneva, 13-16 February '07 - Closed September 2007

- Explored mechanisms that allow different IdM frameworks to interoperate together
- Four deliverables sent to relevant Study Groups and to ISO/IEC/JTC 1/SC27
- Work will continue under banner IdM-GSI

■ **From/In/To Cars Communication (I and II)**

Meeting 1: Geneva, 26 Jan '07

- Working on specifications that will enhance communications in vehicles
- Recently expanded to include wideband communications

■ **IPTV** (see following example)

Meeting 1: Geneva 10-14 July '06

- Coordinating and promoting the development of global IPTV standards

ITU-T Focus Groups (2)

■ **Security Baseline for Network Operators**

Meeting 1: 30 March 2006 - Closed September 2007

- Defined a security baseline so that operators can assess their network and information security posture in terms of available standards
- Output will be published as supplement to X.800 – Security Architecture

■ **NGN Management**

Established Feb 2005

- Providing a centralized approach regarding specification of NGN related fault, configuration, accounting, performance, and security management interfaces

■ **Open Communications Architecture Forum**

Established May 2004

- Defining specifications for a set of components for a new carrier grade open platforms

IPTV FG Major achievements (1)

- Collaboration With Other SDOs
 - Liaison relationship with almost all SDOs involved with the development of IPTV
 - Input from ATIS IIF, DSL Forum and Home Gateway Initiative (HGI)
 - Working relationship with DVB project and ETSI TISPAN
- Regional/national bodies and SDOs benefit from contributing to FG IPTV and having their work incorporated into globally recognized and accepted standards

IPTV FG Major achievements (2)

- Agreed to a unique functional architecture covering legacy and NGN (IMS and non-IMS) based platforms
 - Facilitates alignment between other SDOs
- In depth discussion on home network architecture and other related issues
- Captured vast number of requirements covering home network, metadata, middleware, public interest, QoS/QoE, security, unicast and multicast

Cooperation

MoU between ITU and ETSI

■ Considering:

- ITU as the intergovernmental and universal organization responsible for worldwide standardization of telecommunications
- ETSI as a regional organization

■ Allows

- Each organization to use the documents of the other in order to advance its own work
- Each will also be able to reference standards texts of the other
- Each organizations experts can participate in the others meetings

ITU-T and 3GPP

- ITU-T participated at 3GPP PCG/OP meeting in New Delhi, April 2007
- ITU-T will participate at future 3GPP PCG/OP meetings
- 3GPP "recognize[s] the importance of the relationship between 3GPP and the ITU"
- Input from ITU-T into 3GPP:
 - ITU-T can submit requirements on common IMS either through
 - ▶ a liaison statement, or
 - ▶ through an individual 3GPP membership contribution
- Output from 3GPP into ITU-T: specs on common IMS to be submitted to ITU-T
- September, 2007 meeting TSB and 3GPP management
 - ITU-T invitation to host 3GPP meetings
 - 3GPP as an ITU-T Focus Group?

ITU-T and IEEE

- Feb 2007: Mutual promotion of events of common interest was agreed in February 2007
- May 2007: IEEE Global Standards and Developing Economies: Broadband Access and Infrastructure Seminar, Tunis May 9-10. ITU-T participation in IEEE-SA Corporate Advisory Group (CAG)
- May 2007: IEEE 802.1 and 802.3 Interim Meeting hosted by ITU-T in Geneva May 28-31
- May 2007: This Joint ITU-T/IEEE Workshop on Carrier-class Ethernet in Geneva May 31-June 1
- June 2007: ITU-T welcomes IEEE 802 participants to its 4-15 June SG15

ITU-T and IETF

- ITU-T Workshop on NGN in collaboration with IETF
Geneva, 1-2 May 2005
 - High-level management meeting
- ITU-T / IETF Leadership Gathering
21 July 2007

Global Standards Collaboration

- GSC 12, Kobe 8-13 July 2007
- The mandate of GSC is to provide a venue for the leaders of the Participating Standards Organizations and the ITU to:
 - ...exchange information on the progress of standards development...
 - Collaborate in planning future standards development to gain synergy and to reduce duplication.
 - ...the mandate of GSC is to provide a venue... to:
 - Support the ITU as the preeminent global telecommunication and radiocommunication standards development organization.

WSC Members have aligned their patent policies

- WSC = World Standards Cooperation (ISO, IEC and ITU)
- Based on ITU-T policy
- ... strongly encourages the disclosure of known patented technology from the outset.
- Allows for companies' IPR to be included in standards as long as it is made available under reasonable and non-discriminatory terms and conditions.
- WSC also adopted Guidelines for the Implementation of the Common Patent Policy and a Patent Statement and License Declaration Form.
- Each of the three WSC organizations also has an online patent database.

Global Standards Symposium

- To be held day before next ITU World Telecommunication Standardization Assembly (WTSA), October 2008
- Aims to bring together all standards makers
- Themes:
 - Reducing the standardization gap
 - Improving collaboration
- Details discussed at: tsagroundtable@itu.int

World Telecommunication Standardization Assembly October 2008

- WTSA sets direction and structure for ITU-T
- Meets every four years
- Defines the general policy for the Sector
- Establishes the study group structure
 - Correspondence Group on SG Structure:
tsagstructure@itu.int
- Approves their expected work programme
- Appoints their chairmen and vice-chairmen
- WTSA Resolutions are available at:
www.itu.int/ITU-T/WTSA/resolutions

Free Recommendations

- Historic decision taken by the 2007 Council
- In 2001, Council approved three free Recs per year
- In 2006, Council approved a trial programme of free ITU-T Recommendations online, without restriction
- In 2007, Council made this trial permanent

ITU-T Recommendations	2006	2007	Change
Downloaded, Free of charge, Jan-April	13,601	968,790	+7,100%
Sold, on paper and via e-bookshop, Jan-April	2,532	436	-83%



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Back up slides

Work highlights 2006

- GPON Interoperability demonstrations
- VDSL 2
- G.9954 - Home networking transceivers
- Allocation of intl. dialling code 888 to UN
- Recs J.360-366 – IPCablecom 2
- 14 new signalling protocols
- Architecture of Transport MPLS (T-MPLS)
- New work area – generic sound activity detection
- Accessibility checklist
- Rec. H.324 – speeds up set up time for 3G calls
- Rec. G.729.1 – scaleable voice codec
- ICT Security standards roadmap
- Global NGN standards progressing rapidly within ITU-T
- Providing global leadership through overall framework and structure
- Building on the work of other bodies (not reinventing ...)
- 22 Recommendations on NGN
 - Transition from legacy, requirements, architecture, security, QoS, OAM (Ethernet and MPLS)

Forums/Consortia Approved for Cooperation and Exchange of information

- G Association
- ASN.1 Consortium
- ATSC (Advanced Television Systems Committee)
- AVS (Audio-Visual coding Standard working group)
- DSL Forum
- ETIS (e-and telecommunication information services)
- GSM Association
- Home Gateway Initiative
- IMTC (Multimedia)
- IPDR Organization
- IPsphere Forum
- IPv6 Forum
- Liberty Alliance
- MEF (Metro Ethernet Forum)
- MMTA (Mobile Multimedia Technology Alliance)
- MFA Forum
- MSF (Multiservice Switching Forum)
- NRO (Number Resource Organization)
- OASIS
- OIF (Optical Internetworking Forum)
- OMA (Open Mobile Alliance)
- OMG (Object Management Group)
- SDL Forum Society
- SDL Task Force Consortium
- TM Forum (Tele Management Forum)
- UNICODE
- W3C (World Wide Web Consortium)

Under study: CEPCA (Consumer Electronics Powerline Communication Alliance), EPC Global, MAAWG, UPnP Forum



SDOs Approved for Cooperation and Exchange of Information

- ARIB (Association of Radio Industries and Businesses)
- ATIS (Alliance for Telecommunications Industry Solutions)
- CEA (Consumer Electronics Association)
- CCSA (China Communications Standards Association)
- Ecma International
- ETSI (European Telecommunications Standards Institute)
- IEEE (Institute of Electrical and Electronics Engineers)
- JCTEA (Japan Cable Television Engineering Association)
- NIST (National Institute of Standards and Technology)
- SCTE (Society of Cable Telecommunications Engineers)
- SMPTE (Society of Motion Picture and Television Engineers)
- TIA (Telecommunications Industry Association)
- TTA (Telecommunications Technology Association)
- TC (Telecommunication Technology Committee)

Organizations Qualified for Including References in ITU-T Recommendations

- ARIB (Association of Radio Industries and Businesses)
- ATIS (Alliance for Telecommunications Industry Solutions)
- ATSC (Advanced Television Systems Committee)
- AVS (Audio-Visual coding Standard working group)
- CCSA (China Communications Standards Association)
- DSL Forum
- Ecma International
- EPC Global
- ETSI (European Telecommunications Standards Institute)
- Home Gateway Initiative
- IEEE (Institute of Electrical and Electronics Engineers)
- IMTC (Multimedia)
- ISOC/IETF (Internet Society/Internet Engineering Task Force)
- JCTEA (Japan Cable Television Engineering Association)
- Liberty Alliance
- MAAWG
- MEF (Metro Ethernet Forum)
- MFA Forum
- NIST (National Institute of Standards and Technology)
- OASIS
- OIF (Optical Internetworking Forum)
- OMA (Open Mobile Alliance)
- OMG (Object Management Group)
- SCTE (Society of Cable Telecommunications Engineers)
- SMPTE (Society of Motion Picture and Television Engineers)
- TIA (Telecommunications Industry Association)
- TM Forum (Tele Management Forum)
- TTA (Telecommunications Technology Association)
- TTC (Telecommunication Technology Committee)
- UNICODE
- UPnP Forum
- W3C (World Wide Web Consortium)

Under study: EPC Global, MAAWG, UPnP Forum

